

Dangerous Flowers (antennas)

After yet another fight with his girlfriend, William knows very well that there is only one way to be forgiven: bring her a bouquet of fresh *edelweiss* flowers, her favourites. These flowers can be easily found in the mountains where William lives, however, you are not allowed to pick them unless you have a special permit. But there's no time for bureaucracy after such a fight, so he gotta take the risk and pick them without it!



Figure 1: *Leontopodium nivale*, commonly known as edelweiss.

Unfortunately, after picking up the very first flower, he triggered an unexpected alarm! By quickly hacking the alarm control unit, William realised that the warning is going to be transmitted to the forestry police by a line of N directional antennas in the mountains. Each antenna has a receiver with a threshold level of L_i decibel, and is able to transmit to following antennas with a power of P_i decibels. The signal decreases in strength by D decibel while travelling the distance between any two consecutive antennas. For example, if we have the following configuration of $N = 4$ antennas with a decay $D = 4$:

L:	5	6	3	0
P:	10	7	5	8
i :	0	1	2	3

In this case, the signal from antenna 0 is received by antenna 1 (where arrives with strength 6), but not from the subsequent ones (where arrives with strength 2 and -2 respectively); the signal from antenna 1 is received by antenna 2; and the signal from antenna 2 is received by antenna 3. With the same configuration but a decay $D = 3$, the signal from antenna 1 would have been received by all the following ones, and similarly for the others.

Since the antennas need to run with a very low energy consumption, they are always receiving signals, but are transmitting only at regular intervals of T_i milliseconds starting from S_i : thus, the transmitting times are S_i , $S_i + T_i$, $S_i + 2T_i$ and so on. If an antenna receives a signal at a certain time t , it is able to transmit it at its first transmitting times **strictly after** t .

Help William by computing when the warning is going to be received by the forestry police (which listens directly on the antenna $N - 1$) so he can plan when to run away with the flowers he got!

📎 Among the attachments of this task you may find a template file `antennas.*` with a sample incomplete implementation.

Input

The first line contains the two integers N , D . The following N lines contains four integers L_i , P_i , S_i , T_i .

Output

You need to write a single line with an integer: the earliest time in which the last antenna will transmit the warning. If no such time exist (the warning will not be received) then you should write -1 .

Constraints

- $2 \leq N \leq 100\,000$.
- $0 \leq D \leq 10\,000$.
- $-10^9 \leq L_i < P_i \leq 10^9$ for each $i = 0 \dots N - 1$.
- $0 \leq S_i < T_i \leq 10\,000$ for each $i = 0 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

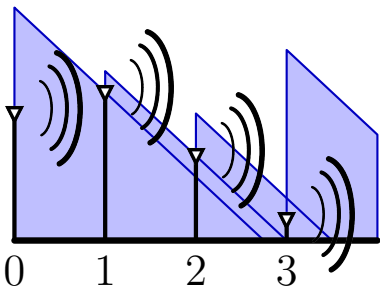
- | | |
|---|---|
| – Subtask 1 (0 points) | Examples. |
|  | |
| – Subtask 2 (10 points) | $N = 2$. |
|  | |
| – Subtask 3 (10 points) | $P_i = 10\,000$, $L_i = 0$ for all i and $D = 10\,000$. |
|  | |
| – Subtask 4 (10 points) | $P_i = 10\,000$, $L_i = 0$ for all i . |
|  | |
| – Subtask 5 (20 points) | $P_i = 10\,000$ for all i . |
|  | |
| – Subtask 6 (25 points) | $N \leq 500$. |
|  | |
| – Subtask 7 (15 points) | $D = 0$. |
|  | |
| – Subtask 8 (10 points) | No additional limitations. |
|  | |

Examples

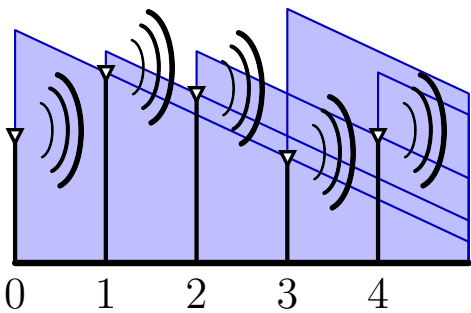
input	output
4 4 5 10 5 20 6 7 2 10 3 5 12 15 0 8 0 1	28
5 2 5 10 2 4 8 9 5 15 7 9 1 7 4 11 2 20 5 8 4 6	10
5 4 2 8 4 7 5 9 2 3 0 7 3 8 5 8 5 9 1 5 1 6	-1

Explanation

In the **first sample case**, antennas are configured as explained in the task statement (see also the picture below, where the height of the antenna represents its threshold, and the blue triangle represents the area where the power is enough). Every antenna transmits to the following one at time 5, 12, 27 and 28 respectively.



In the **second sample case**, antenna 0 is received by antennas 1 and 3; antenna 1 is received by antennas 2 and 3; antenna 2 is received by antennas 3 and 4; antenna 3 is received by antenna 4. Antennas transmit the warning for the first time respectively at 2, 5, 8, 22, 10.



In the **third sample case**, antennas transmit for the first time respectively at 4, never, 11, never, never. Notice that the fourth antenna could be able to communicate with the fifth, but it cannot receive from any of the previous, so the warning is not received and the answer is -1 .

